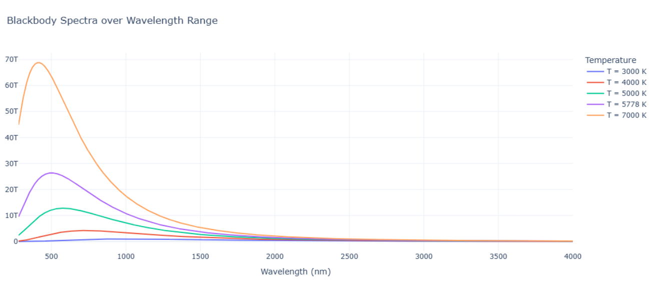
★ HOMEWORK

(a)

$$B_\nu(\nu, T) = \left[\frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_bT)} - 1}\right]$$

$$B_\lambda(\lambda, T) = \left[\frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_bT)} - 1}\right]$$



BB spectrum for diff. temperatures over the wavelength cosidered in astm1.5G spectrum

BB spectrum for different temperatures for corresponding frequencies

(b)

$$\int B_\lambda \, d\lambda = \int \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_bT)} - 1} \, d\lambda$$

$$= 2hc^2 \int \frac{1}{\lambda^5} \frac{1}{e^{hc/(\lambda k_bT)} - 1} \, d\lambda$$

$$x = \frac{hc}{\lambda k_b T} \; ; \; \frac{dx}{d\lambda} = \frac{-hc}{\lambda^2 k_b T}$$

$$\Rightarrow x^3 dx = \frac{-h^4 c^4}{\lambda^5 k_b^4 T^4} d\lambda$$

$$= 2hc^2 \int \frac{1}{\cancel{\lambda^5}} \frac{1}{e^x - 1} \frac{\cancel{\lambda^5} k_b^4 T^4}{-h^4 c^4} x^3 dx$$

$$= -\frac{2 k_b^4 T^4}{h^3 c^2} \underbrace{\int_0^\infty \frac{x^3}{e^x - 1} dx}_{\frac{\pi^4}{15}}$$

$$\boxed{|P| = \frac{2 k_b^4 \pi^4}{h^3 c^2 15} T^4}$$

(c) $$B_\nu(\nu,T) = \left[\frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/(k_bT)}-1}\right]$$

$$\int B_\nu\, d\nu = \int \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/(k_bT)}-1}\, d\nu$$

$$x = \frac{h\nu}{k_bT} \; ; \; \frac{dx}{d\nu} = \frac{h}{k_bT}$$

$$= \frac{2k_b^3T^3}{h^2c^2}\int \frac{h^3\nu^3}{k_b^3T^3}\frac{1}{e^x-1}\frac{k_bT}{h}\, dx$$

$$= \frac{2k_b^4T^4}{h^3c^2}\underbrace{\int_0^\infty \frac{x^3}{e^x-1}\, dx}_{\frac{\pi^4}{15}}$$

$$\boxed{P = \frac{2k_b^4\pi^4}{h^3c^2 15}T^4}$$

(d) Yes, the power expression matches from λ & ν

terms.

(e) No. of photons $(n_\lambda(T)) = \dfrac{\int B_\lambda(T)}{E} = \dfrac{\int B_\lambda(T)}{\frac{hc}{\lambda}}$

$$B_\lambda(\lambda, T) = \left[\frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_b T)} - 1}\right]$$

$$\int_0^\infty n_\lambda(T) = \int_0^\infty \left[\frac{2c}{\lambda^4} \frac{1}{e^{hc/(\lambda k_b T)} - 1}\right] d\lambda$$

$$x = \frac{hc}{\lambda k_b T} \; ; \; x^2 dx = \frac{-h^3c^3}{\lambda^4 k_b^3 T^3} d\lambda$$

$$= -2c \int_0^\infty \frac{1}{\cancel{x^4}} \frac{1}{e^x - 1} \frac{\cancel{x^4} k_b^3 T^3}{h^3 c^3} x^2 dx$$

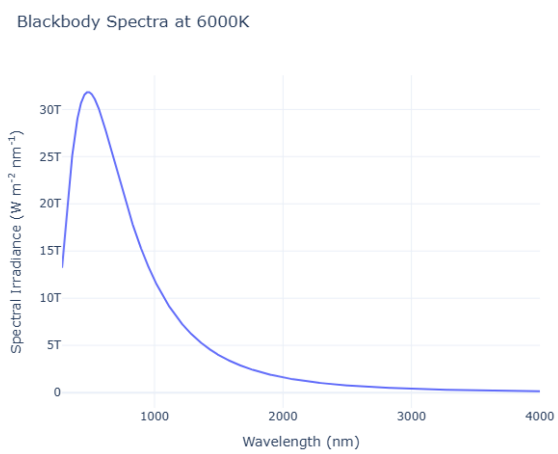
Total no. of photons $= \dfrac{-2 k_b^3 T^3}{h^3 c^2} \underbrace{\displaystyle\int_0^\infty \frac{x^2}{e^x - 1} dx}_{2\zeta(3) = 2.40411}$

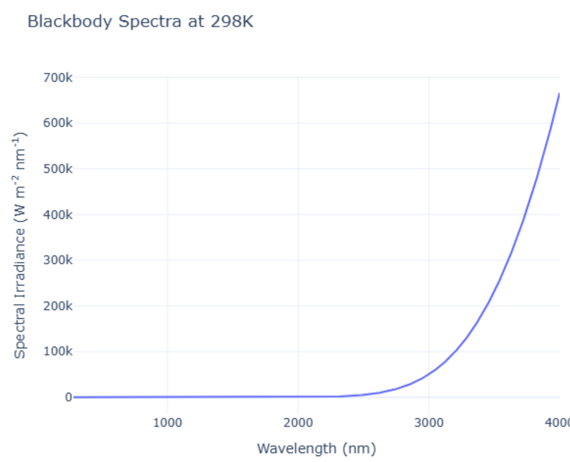
$$\text{No. of photons} = \frac{4.80822\, k_b^3\, T^3}{h^3 c^2}$$

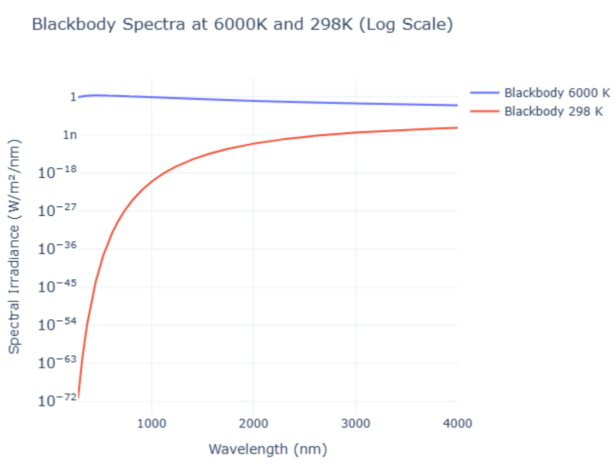
$$\therefore \text{No. of photons} \propto T^3$$

Assignment - 2

(A)







Plotted for $\lambda \sim$ (280nm to 4000nm)

(B) According to Planck's law,

$$B_\lambda(\lambda, T) = \left[\frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_b T)} - 1}\right]$$

$W\,m^{-3}\,sr^{-1}$

But AM1.5G is E_λ which has units of $W\,m^{-2}\,nm^{-1}$

To convert B_λ to E_λ, we multiply it with solar solid angle,

$$\Omega_{sun} = \pi \left(\frac{R_{sun}}{d_{sun}}\right)^2$$

where, R_{sun} = Radius of sun

d_{sun} = Distance betn. sun & earth.

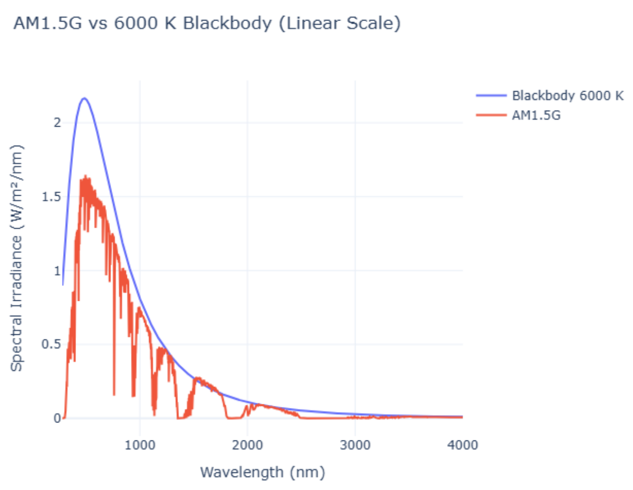
$$E_\lambda(\lambda, T) = B_\lambda(\lambda, T) \cdot \Omega_{sun} \cdot 10^{-9}$$

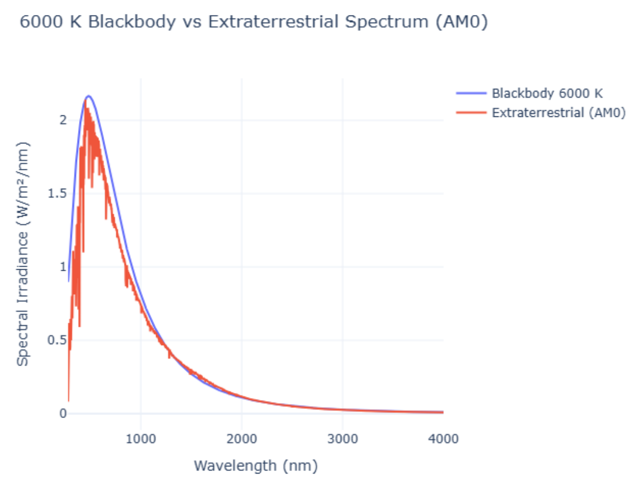
$(W\,m^{-2}\,nm^{-1})$

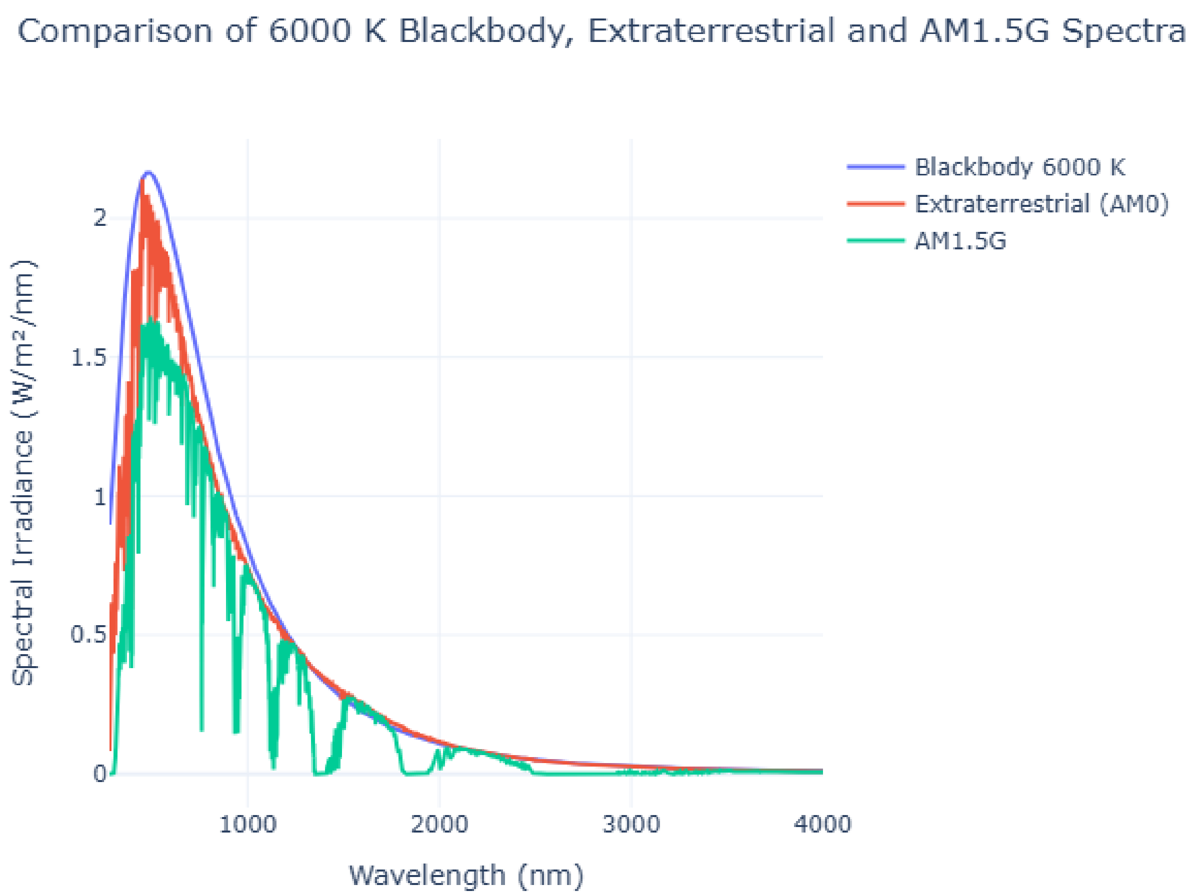
The geometrical factor f_ω is dependent upon the solid angle subtended by the sun and the angle of incidence upon the solar battery. The solid angle subtended by the sun is denoted by ω_s, where

$$\omega_s = \pi(D/L)^2/4 = \pi(1.39/149)^2/4 = 6.85\times10^{-5}\text{ sr}, \tag{3.3}$$

and D, L are, respectively, the diameter and distance of the sun, taken as 1.39 and 149 million km.







Plotted for $\lambda \sim$ (280nm to 4000nm)

ϕ_{BB} and $\phi_{AM1.5G}$ do not overlap and instead show a lot of variance.
This is because, sun is a perfect black body, whereas AM1.5G is measured irradiance after passing through atmosphere.
The dips occur due to irradiance getting absorbed by Ozone layer, Water vapour and other gases.

ϕ_{BB} and ϕ_{AM0G} overlap. Only difference is in the magnitude of flux. This is because ϕ_{BB} is measured @ 6000 K whereas ϕ_{AM0G} is measured @ 298 K (This can be seen in the log graph shown above).

(c) $J = -J_{sc} + q\,(RR_0)\left(e^{qV/k_bT} - 1\right)$

$$RR_0 = \int_{E_g}^{\infty} \frac{\phi_{BB}}{E_{ph}}\, d\lambda$$

$$E_g = \frac{1240}{\lambda(\text{in nm})}$$

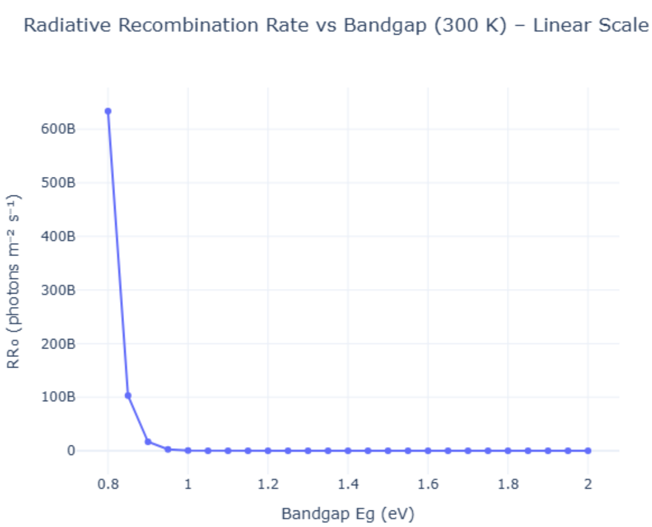
From HOMEWORK Part (e),

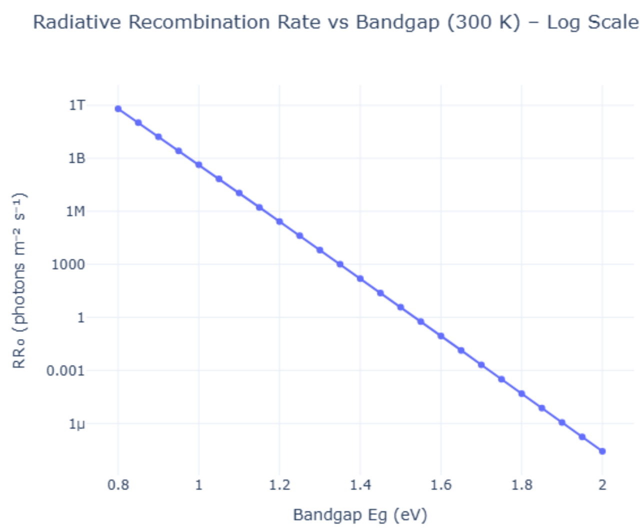
$$\eta_\lambda(T) = \int_{E_g}^{\infty} \left[\frac{-2c}{\lambda^4} \frac{1}{e^{hc/(\lambda k_b T)} - 1} \right] d\lambda$$

OR $$RR_0(E_g) = \int_{\infty}^{E_g} \frac{2c}{\lambda^4} \frac{1}{e^{hc/k_bT} - 1}\, d\lambda$$

$E_g = \frac{1240}{\lambda} \Rightarrow$ Lower Limit $= 0$
Upper Limit $= \frac{1240}{\lambda}$

$$RR_0(E_g) = \int_0^{\frac{1240}{\lambda}} \frac{2c}{\lambda^4} \frac{1}{e^{hc/\lambda k_b T} - 1} \, d\lambda$$





For $E_g = 1.5\,eV \Rightarrow \lambda = \frac{1240}{1.5} = 826.67$

$$RR_0(1.5) = \int_0^{826.67} \frac{2c}{\lambda^4} \frac{1}{e^{hc/\lambda k_b T} - 1} \, d\lambda$$

$$= 3.717 \text{ photons } m^{-2} s^{-1}$$

$$RR_0(E_g) \propto e^{-E_g/k_b T}$$

(D) $J = -J_{sc} + q(RR_0)\left(e^{qV/k_bT} - 1\right)$

$$\boxed{J_{sc}(E_g) = q\int_0^{\frac{1240}{\lambda}} \frac{\Phi_{AM1.5G}}{E}\, d\lambda}$$

$$\boxed{q\,RR_0(E_g) = q\int_0^{\frac{1240}{\lambda}} \frac{2c}{\lambda^4}\frac{1}{e^{hc/\lambda k_bT} - 1}\, d\lambda}$$

For V_{oc} ; $J = 0$

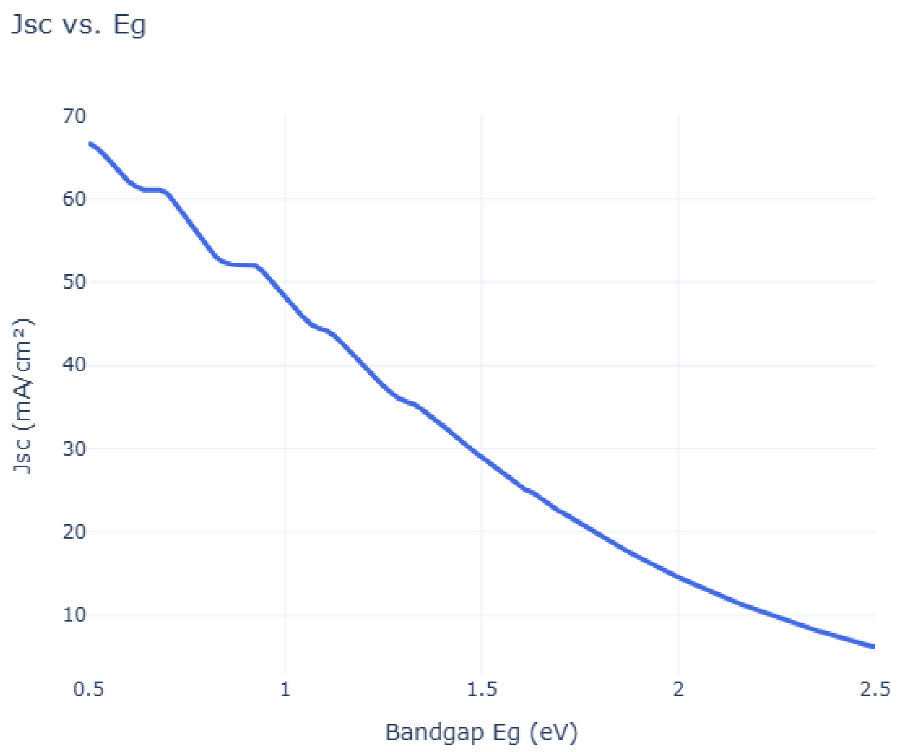
$$J_{sc} = q(RR_0)\left(e^{qV/k_bT} - 1\right)$$

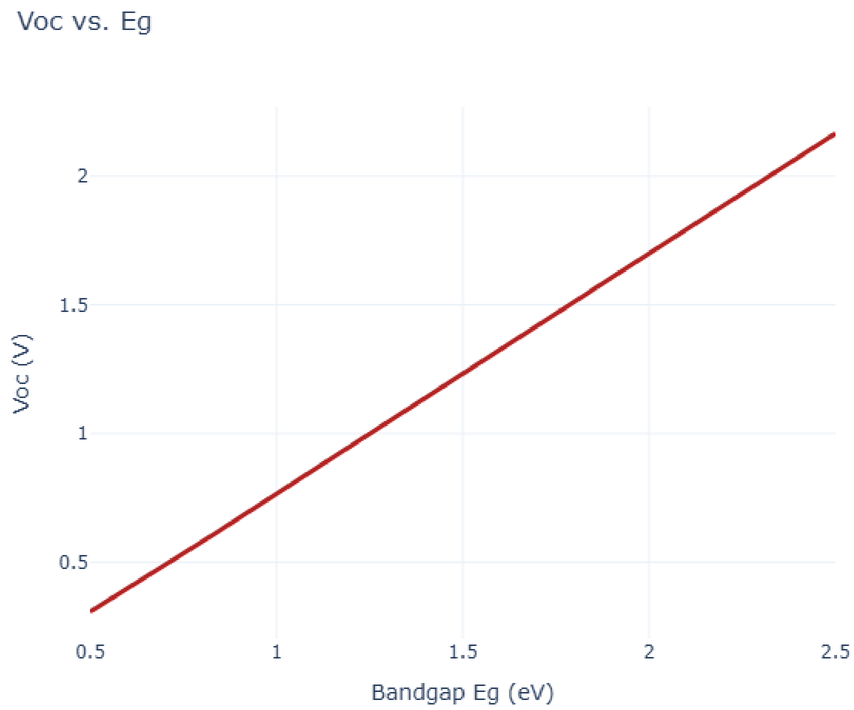
$$\frac{J_{sc}}{q(RR_0)} + 1 = e^{qV/k_bT}$$

$$\boxed{V_{oc} = \frac{k_bT}{q}\ln\left(\frac{J_{sc}}{q(RR_0)} + 1\right)}$$

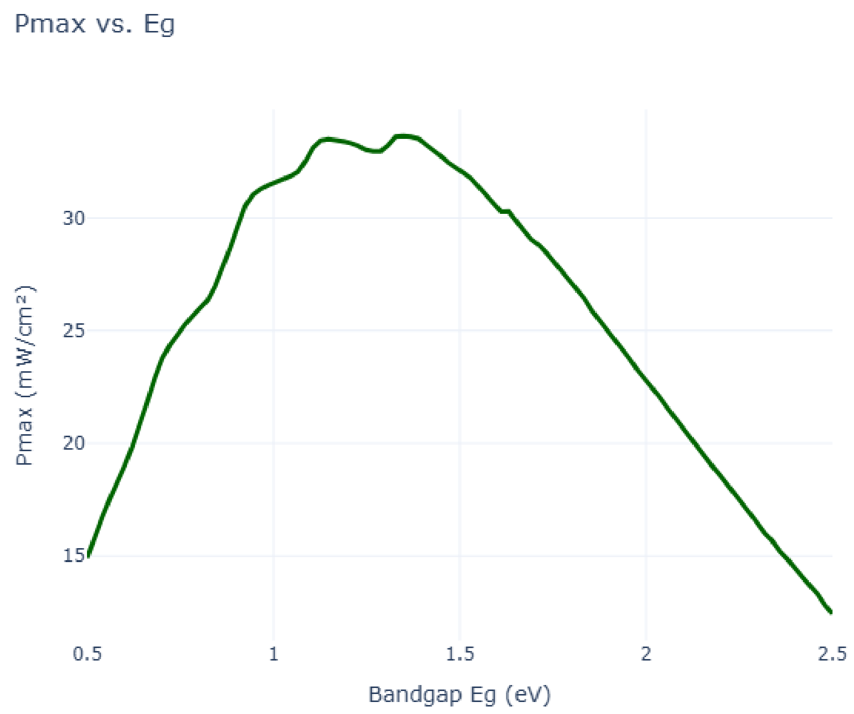
$$P(V) = J(V)V$$

$$= \left[q\int_0^{\frac{1240}{\lambda}} \frac{\Phi_{AM1.5G}}{E}\, d\lambda - q\int_0^{\frac{1240}{\lambda}} \frac{2c}{\lambda^4} \frac{1}{e^{hc/\lambda k_b T} - 1}\, d\lambda \right] V$$



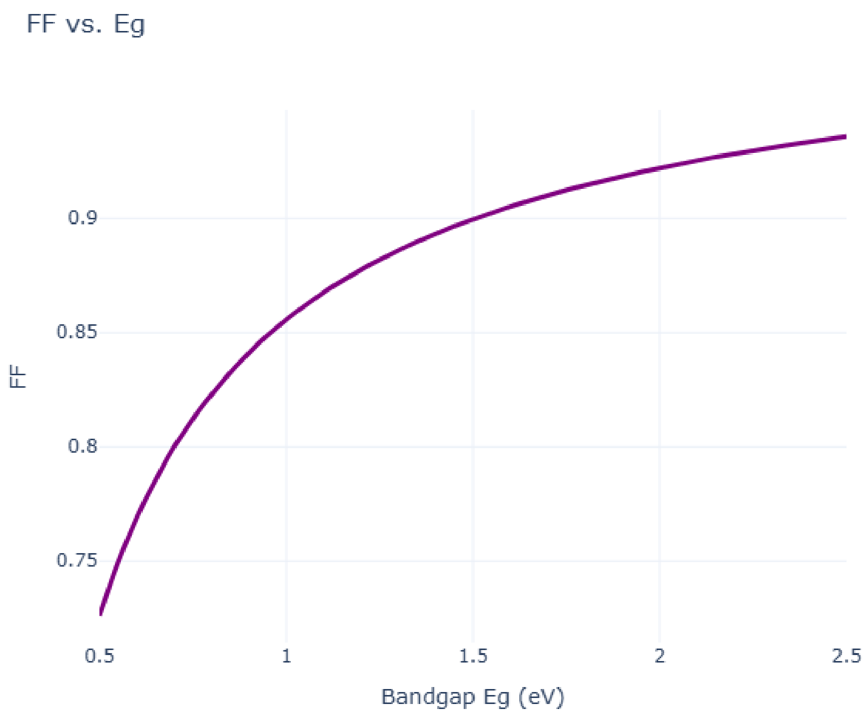


Voc vs. Eg
Voc (V)
2
1.5
1
0.5
0.5
1
1.5
2
2.5
Bandgap Eg (eV)

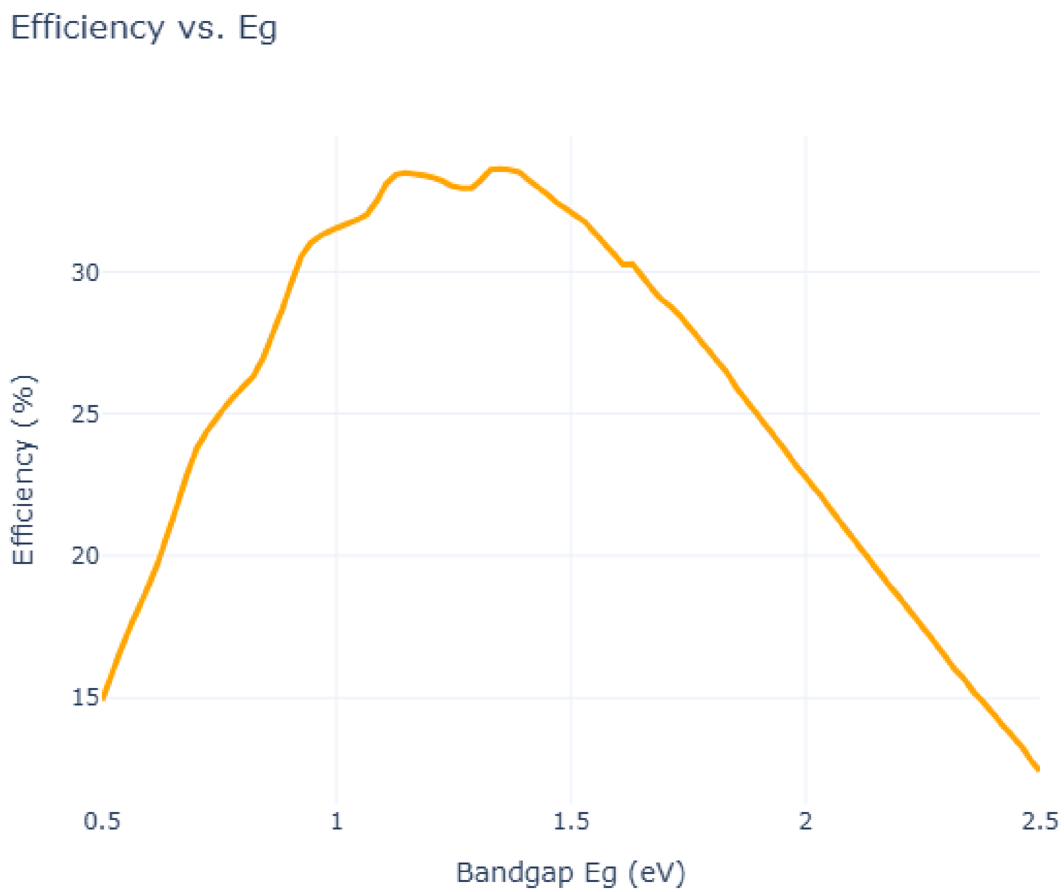


Pmax vs. Eg
Pmax (mW/cm²)
30
25
20
15
0.5
1
1.5
2
2.5
Bandgap Eg (eV)

$$FF = \frac{P_{max}}{V_{oc} \cdot J_{sc}}$$



$$\eta = \frac{P_{max}}{P_{incident}}$$



Highest Efficiency = 33.62% @ $E_g = 1.3485$ eV